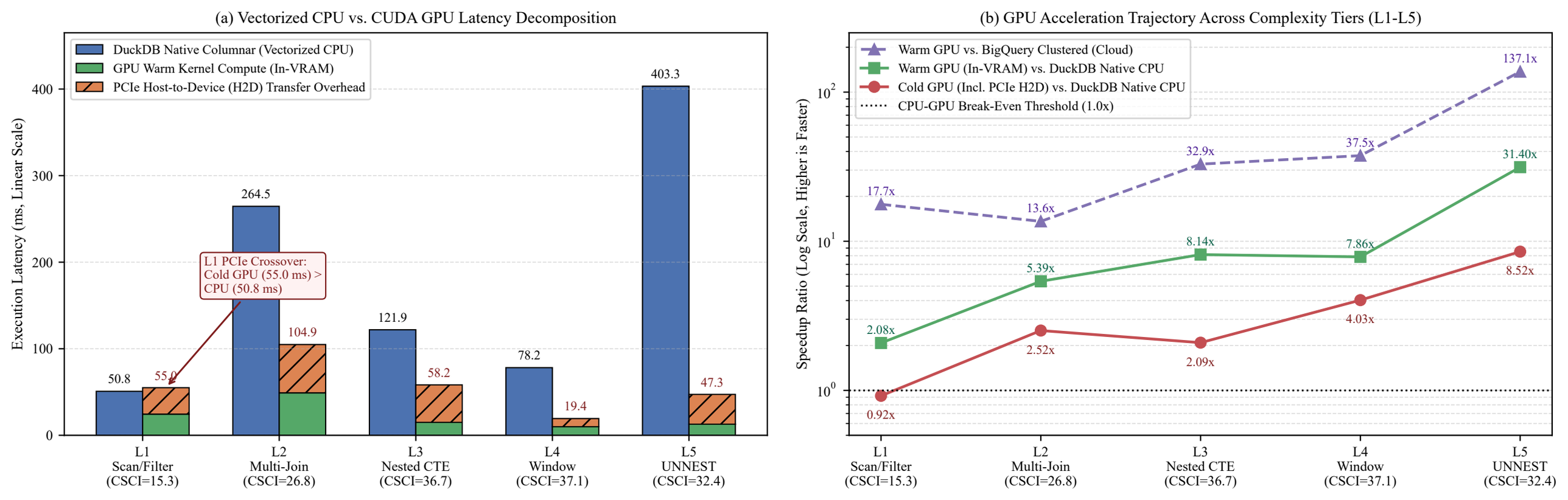
(a) Vectorized CPU vs. CUDA GPU Latency Decomposition
DuckDB Native Columnar (Vectorized CPU)
GPU Warm Kernel Compute (In-VRAM)
PCIe Host-to-Device (H2D) Transfer Overhead
Execution Latency (ms, Linear Scale)
0
100
200
300
400
50.8
55.0
264.5
104.9
121.9
58.2
78.2
19.4
403.3
47.3
L1 PCIe Crossover:
Cold GPU (55.0 ms) >
CPU (50.8 ms)
L1
Scan/Filter
(CSCI=15.3)
L2
Multi-Join
(CSCI=26.8)
L3
Nested CTE
(CSCI=36.7)
L4
Window
(CSCI=37.1)
L5
UNNEST
(CSCI=32.4)
(b) GPU Acceleration Trajectory Across Complexity Tiers (L1-L5)
Warm GPU vs. BigQuery Clustered (Cloud)
Warm GPU (In-VRAM) vs. DuckDB Native CPU
Cold GPU (Incl. PCIe H2D) vs. DuckDB Native CPU
CPU-GPU Break-Even Threshold (1.0x)
Speedup Ratio (Log Scale, Higher is Faster)
$10^0$
$10^1$
$10^2$
17.7x
13.6x
32.9x
37.5x
137.1x
2.08x
5.39x
8.14x
7.86x
31.40x
0.92x
2.52x
2.09x
4.03x
8.52x
L1
Scan/Filter
(CSCI=15.3)
L2
Multi-Join
(CSCI=26.8)
L3
Nested CTE
(CSCI=36.7)
L4
Window
(CSCI=37.1)
L5
UNNEST
(CSCI=32.4)